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This assignment has 10 questions of 8 Marks each, answer all questions. Rest 20 marks are for viva voce. You may use illustrations and diagrams to enhance the explanations. Please go through the guidelines regarding assignments given in the Programme Guide for the format of presentation.

- Q1:** Define the term data science. Describe its applications in two industries of your choice (e.g., healthcare, finance, e-commerce). What role does the data science lifecycle play in managing data projects?
- Q2:** Explain Exploratory Data Analysis (EDA) and its importance. What are the main steps in performing EDA on a new dataset? Describe two methods for detecting outliers and how handling outliers impacts data analysis.
- Q3:** Describe the role of statistical hypothesis testing in data analysis. What are Type I and Type II errors, and how do they affect decision-making? Provide an example of hypothesis testing in a real-world scenario.
- Q4:** Discuss the 4 Vs of big data (Volume, Velocity, Variety, and Veracity). Provide a real-world example of each, explaining how these characteristics create challenges in big data management.
- Q5:** Explain the Hadoop architecture with a focus on HDFS and the master/slave architecture. How do NameNode and DataNodes work together to store and manage large datasets? Provide a basic example of this storage process.
- Q6:** Compare Apache Spark, Hive, and HBase in terms of functionality, data processing methods, and use cases. When would Spark be preferred over traditional MapReduce, and why?
- Q7:** Describe the purpose and functionality of a *Bloom filter* in data stream processing. How does the Bloom filter efficiently check for element presence? Describe the Flajolet-Martin algorithm for cardinality estimation in data streams.
- Q8:** What is the PageRank algorithm, and how is it used in link analysis? Describe the concept of “flow of rank” in PageRank. Explain how the PageRank of a web page is calculated using the flow model.
- Q9:** Discuss the challenges of online advertising and recommendation systems. Explain the concept of collaborative filtering with an example, and discuss the role of clustering in social network analysis.
- Q10:** What is the Random Forest algorithm? Explain how it can be applied to classification problems. Write a program in R to implement a Random Forest classifier on a sample dataset and explain its output.



MCS-226

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MCAOL (III) Semester.

Q. Define the term data Science. Describe its application in two industries of your choice (e.g. healthcare, finance, e-commerce). What role does the data Science Lifecycle play in managing data projects?

Ans. what is Data Science?

It is all about making sense of vast amounts of data. It combines method from statistics, programming, machine learning, and data visualization to unlock hidden patterns, trends, and insights from both structured and unstructured data. The goal is to turn raw data into useful information that can support decision-making in various industries.

Application of Data Science in Healthcare and Finance.

1. Healthcare:-

Data Science is revolutionizing how we treat patients and manage healthcare systems. Here how it works:

- Predictive analytics for Disease Diagnosis:

By analyzing patient data, such as medical history, test results, and lifestyle factors, data science can help predict diseases before they become serious.

- Personalized Treatment Plans:

Every patient is unique, and data science allows healthcare providers to customize treatments based on a person's genetic information, medical history, and even their response to previous treatments.

- Operational Efficiency:-

Healthcare systems can also use data science to improve their operations.

2. Finance:-

Data Science plays a critical role in the financial industry, from improving risk management to detecting fraud and predicting market trends.

• Credit Scoring and Risk Management.

Data Science allows banks and other lenders to analyze a person's financial history.

• Fraud Detection

Financial institutions are increasingly using data science to spot fraudulent activities in real time.

• Algorithmic Trading:-

In the world of investments, data science is helping to predict stock market trends by analyzing large amounts of historical data.

- The Role of the Data Science Lifecycle in Managing Projects:
The data science lifecycle is like a roadmap that guides a data project from start to finish:

• Problem Definition:-

Before diving into the data, you need to clearly understand the problem you're trying to solve. Whether it's improving patient care, predicting customer behavior, or spotting fraud.

• 2. Data Collection:-

After identifying the problem, the next step is gathering data. This could involve pulling data from existing systems, purchasing external datasets, or even collecting new data through sensors, surveys, or online platforms.

3. Data Cleaning:-

Raw data is rarely perfect. It often has missing values, duplicates, or errors. This is all about cleaning the data to make it



4. Exploratory Data analysis (EDA)
with clean data in hand, data scientists explore the dataset to understand its structure and discover any underlying trends or patterns.

5. Model Building and Evaluation:-

Data Science doesn't stop at deployment. Once a model is in use, it needs to be continuously monitored to ensure it says. These models use algorithms that learn from the data to make prediction or classification.

6. Model deployment:-

Once a model is proven to work well, it's deployed in the real world. This means integrating it into business process or making it available for use by stakeholders.

7. Monitoring and Maintenance:

Data science doesn't stop at deployment. Once a model is in use, it needs to be continuously monitored to ensure it available for use by stakeholders. For ex.



Q2 Explain Exploratory data analysis (EDA) and its importance. What are the main steps in performing EDA on a new dataset. Describe two methods for detecting outliers and how handling outliers impacts data analysis.

Ans:- A group of methods that can be used to learn possibilities of relationships amongst data. After you have obtained relevant data for analysis instead of performing the final analysis, you may like to explore the data for possible relationship using exploratory data analysis.

1. As a first step, you may perform the description of various categorical and qualitative variable of your data.

a. for the qualitative data, you may create frequency tables and bar charts to know the distribution of data among different categories.

b. you may compute the mean, median, standard deviation, skewness and kurtosis.

The kurtosis value relates to peaks in the data.

2. After performing the univariate analysis, you may try to perform some bi-variate analysis. Some of the basic statistics.

3. Finally, you may like to look into the possibilities of multi-variate relationships amongst data. you may use dimensionality reduction by using techniques feature extraction or principal component analysis, you may perform clustering to identify possible set of classes in the solution space or you may use graphed tools.



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= Two methods for detecting outliers:-

Outliers are data points that differ significantly from most other values in the dataset. They can mess with your analysis and models.

1. Z-Score Method:-

A Z-score measures how far a data point is from the mean in terms of standard deviations. If a Z-score is greater than 3 or less than -3, the

2. IQR (Interquartile Range):- The IQR method is based on the spread of the middle 50% of the data.

Q3:- Describe the role of statistical hypothesis testing in data analysis.

SI. To determine statistical analysis is mainly used in the two -
- Statistical hypothesis testing is a key technique in data analysis that helps us make informed decisions based on data. It's like a structured way to test assumptions or claims we might have about a dataset.

- How Hypothesis Testing works:-

1. State your hypothesis.

- Null Hypothesis (H_0):- This is the "default" assumption that there is no effect or difference (e.g. the new drug has no effect).
- Alternative Hypothesis (H_1)
This is the hypothesis we are trying to test suggesting that there is an effect or difference (e.g. the new drug does have an effect).

2. Set the significance level (α):

This is the threshold for how much evidence you need to reject the null hypothesis. It's often set to 0.05, meaning you are willing to accept a 5% chance of rejecting the null hypothesis when it's actually true.



3. collect data and Conduct the Test:

Based on the type of data you have, you'll perform a specific statistical test (like a t-test or chi-square test) to see if the data supports the alternative hypothesis.

4. Make your decision:-

After calculating the test statistic, you compare it with the critical value or look at the p-value. If the p-value is smaller than α (usually 0.05)

5. Draw conclusions:-

Based on the results, you conclude whether or not the data provides enough evidence to support the alternative hypothesis.

- Type I and Type II Errors.

1. Type I Error (False Positive):

- This happens when you reject the null hypothesis even though it's true.

Essentially, you think you've found an effect when there isn't one.

- The probability of making a Type I error is equal to your significance level (α). So if $\alpha = 0.05$ there's a 5% chance of making a Type I error.

Ex:- Imagine a new drug is tested, and the hypothesis test incorrectly concludes that it works when in reality it doesn't. This could lead to unnecessary treatments.

2. Type II Error (False Negative)

- A Type II error happens when you fail to reject the null hypothesis even though it's false. Essentially, you miss detecting a real effect or difference.



- The probability of making a Type II error is denoted by β , and the power of the test is $1-\beta$ representing the ability to correctly detect an effect when one exists.

Ex: - if the new drug is effective but the test fails to detect it, you would be making a Type II error.

- How Type I and Type II Errors affect decision-making - Both Type I and Type II errors can influence decision-making significantly, especially in areas like healthcare, finance and public policy:-

- Type I Errors:-

This leads to false alarms. In the medical world, a Type I error might mean approving a treatment that doesn't work, wasting resources and potentially causing harm.

- Type II Errors:-

These errors lead to missed opportunities. For instance, if a Type II error occurs in a drug trial, a treatment that could save lives might be dismissed.

- Real-world example of Hypothesis Testing:-

Let's say a company wants to know whether a new marketing campaign increases sales.

- Null Hypothesis (H_0)

The new marketing campaign doesn't affect sales.

- Alternative Hypothesis (H_1):-

The new marketing campaign does increase sales.

The company sets a significance level of 0.05, meaning they are okay with a 5% chance of wrongly concluding that the campaign works and perform a statistical test (like a t-test) to compare the sales before and after the campaign.

If the p-value from the test is less than 0.05, they would reject the null hypothesis and conclude that the campaign works.



Ques:- Discuss the 4 Vs of big data (volume, velocity, variety and veracity). Provide a real-world example of each, explaining how these characteristics create challenges in big data management.

Ans:- The 4 Vs of big data: Volume, Velocity, Variety and Veracity:

Big data is characterized by four key aspects: volume, velocity, variety, and veracity, which present unique challenges in data management.

1. Volume (data size)

Definition:- Volume refers to the massive amount of data generated every second from multiple sources.

Exam:- The India Aadhaar system, with over 1.3 billion enrolled users, generates an enormous volume of biometric and demographic data.

Ch:- Storing, processing, and analyzing such massive datasets require high computational power and efficient data management strategies.

2. Velocity (data speed)

Definition:- Velocity indicates the speed at which data is generated, collected, and processed.

Example:- UPI (Unified Payments Interface) processes millions of real-time transactions daily. In January 2024, UPI recorded over 12 billion transactions in a month.

Challenge:- Managing real-time data streams with minimal latency while ensuring security and fraud detection is a significant challenge.



3. Variety (different data types)

Definition: Variety refers to the different format of data - structured (database), Semi-structured (JSON, XML), and unstructured (image, video, text).

Example: Social media platforms like Twitter and Facebook generate diverse content in multiple Indian languages.

— Integrating and analyzing diverse data types for meaningful insights is complex and requires advanced AI and ML techniques.

4. Veracity (data accuracy and reliability)

Definition: Veracity concerns the trustworthiness and quality of data.

- Ex: Fake news and misinformation on WhatsApp often mislead people.

Challenge: Ensuring data credibility and combating misinformation requires fact-checking mechanisms and regulatory frameworks.

Q5: Explain the Hadoop architecture with a focus on HDFS and the master/slave architecture. How do NameNode and DataNodes work together to store and manage large datasets? Provide a basic ex of the storage process:-

Ans:- HDFS and Master-Slave Mode

Hadoop is an Open-Source framework designed for processing large datasets across a distributed computing environment. It follows a Master-Slave Architecture where Hadoop Distributed File System (HDFS) plays a crucial role in storing massive data.

1. HDFS Overview:-

HDFS is a fault-tolerant, scalable storage system that breaks large dataset into smaller blocks and distributes them across multiple machines.

The default block size is 128MB or 256MB, ensuring efficient storage:-

2. Master-Slave Architecture:-

1. NameNode (Master): Manages the metadata (file locations, block mapping).
2. DataNodes (Slaves): Store actual data blocks and report to the NameNode to confirm their availability:-

3. Working of NameNode and DataNodes

- When a user uploads a file, HDFS splits it into multiple blocks.
- NameNode: assigns block to various DataNodes and keeps a record of their locations.
- DataNodes store these blocks and send periodic "heartbeat" signals to the NameNode to confirm their availability.
- If a DataNode fails, the NameNode redistributes the lost blocks to other active DataNodes.

4. Ex: of Storage Process

Suppose a 1GB file is uploaded. HDFS splits it into 8 blocks (128MB each) and distributes them across multiple DataNodes. The NameNode maintains a mapping of which block is stored in which DataNode, ensuring high availability and fault tolerance.

Thus, Hadoop architecture ensures scalability, reliability, and efficient big data processing, making it ideal for Indian enterprises handling vast amounts of data.



Qs:- Compare Apache Spark, Hive, and HBase in terms of functionality, data processing methods, and use cases, when would Spark be preferred over traditional MapReduce and why?

Ans:- Apache Spark:-

- In memory distributed data processing engine for real-time and batch processing.
 - In-memory computing, DAS execution, supports batch, streaming, ML and graph processing.
 - Real time analytics, Machine Learning, Graph processing, streaming data processing.
- Apache Hive:-
- SQL-based data warehousing for batch processing on Hadoop.
 - Uses MapReduce support SQL-like querying via HiveQL optimized for ETL jobs.
 - Data warehousing, adhoc queries, ETL processing, Business intelligence
- Apache HBase:-
- NoSQL database for real-time read/write access.
 - Stores structured/unstructured data in HDFS, optimized for fast random reads/writes
 - NoSQL applications, Sensor data, real-time log processing, user profile storage.

→ When is Spark preferred over Traditional MapReduce?

1. Speed and Efficiency:- Spark processes data in-memory, which makes it significantly faster than MapReduce, which relies on disk I/O for intermediate data storage.

2. Real-time processing:-

Unlike MapReduce, which is batch-oriented, Spark supports real-time data streaming via Spark Streaming.



3. Ease of Use:-

Spark supports multiple APIs (Java, Scala, Python, R), making it more developer friendly than MapReduce, which requires complex Java-based implementations.

4. Machine Learning and Graph Processing:-

Spark integrates with MLlib and GraphX, enabling machine learning and graph computation, which MapReduce lacks natively.

5. Multi-purpose Workloads:-

Spark supports SQL, ML, graph processing, and batch workloads within a single framework, whereas MapReduce is limited to batch jobs.

Spark is ideal for applications requiring real-time analytics, iterative computation, and fast batch processing, making it a powerful alternative to traditional MapReduce.

Q7 Describe the purpose and functionality of a * Bloom filter * in data stream processing. How does the bloom filter efficiently check for element presence? Describe the Flajolet-Martin algorithm for cardinality estimation in data streams.

Ans:- Bloom Filter in data stream processing:-

A Bloom filter is a space-efficient probabilistic data structure used to test whether an element is present in a set. It is widely used in data stream processing because it provides fast membership checks while using minimal memory. However, a Bloom filter can produce false positives (indicating an element is present when it is not) but never false negatives (it will never miss an existing element).



— How Bloom Filter Works efficiently:

1. Initialization: A bloom filter consists of a bit array of size m initialized to 0 and k position-independent hash functions.
2. Insertion: When an element is inserted, it is hashed k times and the corresponding k positions in the bit array are set to 1.
3. Lookup: To check if an element is present, the same k hash functions are applied. If all k positions are 1, the element is considered present, otherwise, it is definitely not in the set.

— Flajolet-Martin Algorithm for Cardinality Estimation:

is used to estimate the number of unique elements (cardinality) in a data stream using probabilistic techniques.

How FM Algorithm Works:-

1. Hashing: Each incoming element is hashed to a binary value.
2. Trailing Zeros: The algorithm tracks the maximum number of trailing zeros in these hash values.
3. Estimation: If the longest sequence of trailing zeros is R , the estimated cardinality is approximately 2^R .
4. Averaging: Multiple hash functions and averaging are used to reduce estimation error. Both bloom filter and Flajolet-Martin algorithm are crucial in handling big data streams, optimizing memory, and providing efficient approximations for large-scale applications.



Q8:- what is the PageRank algorithm, and how is it used in link analysis? describe the concept of "flow of rank" in PageRank. Explain how the PageRank of a web page is calculated using the flow model.

Ans:- PageRank Algorithm and link analysis:-
PageRank is an algorithm developed by Larry Page and Sergey Brin, the founders of Google, to rank web pages based on their importance. It evaluates the significance of a webpage by analyzing the number and quality of links pointing to it. The core idea is that a webpage is important if many other important pages link to it.

PageRank is widely used in link analysis to determine the relevance and authority of web pages. It helps search engines rank web result based on link structures rather than just keywords.

— Flow of Rank in PageRank —

The concept of the flow of rank represents how PageRank value is distributed among linked web pages.

A webpage rank is divided among all the pages it links to passing authority or "link juice" to them. If a high-ranking page links to another page, a portion of its rank flows to the linked page, boosting its importance.

Ex:- If Page A links to Page B and Page C, its PageRank is split and passed to both. The more links a page has, the more diluted the rank it passes.

Calculation of PageRank Using the Flow Model.

PageRank (PR)

$$PR(A) = (1-d) + d \frac{PR(B)}{L(B)}$$



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where:-

- $PR(A)$ is the PageRank of Page A,
- d is the damping factor (usually 0.85) representing the probability a user continues clicking links
- $PR(B)$ is the PageRank of linking page B,
- $L(B)$ is the number of outbound links from B.

This iterative process continues until PageRank values stabilize, ensuring fair ranking based on link importance and distribution.

Q9:- Discuss the challenges of online advertising and recommendation systems. Explain the concept of collaborative filtering with an example and discuss the role of clustering in Social Network Analysis.

Ans:- Challenges of online advertising and Recommendation Systems.

1. Data Privacy and Security:-

With increasing concerns over user data privacy, regulations like the Personal Data Protection Bill (PDPB) impact how companies collect and use data.

2. AD Fraud:-

Click fraud and bot-generated traffic lead to financial losses for advertisers.

3. Personalization vs. Privacy:-

While users prefer personalized ads, excessive tracking raises ethical concerns.

4. Algorithmic Bias:-

Biased data can lead to unfair recommendations, reinforcing stereotypes or misinformation.

5. Ad Blocking:-

Many users install ad blockers, reducing ad effectiveness.

6. User Fatigue:-

Constant exposure to ads can cause irritation, leading to lower engagement.



7. Diverse Demographics:-

India multilingual and Socio-economic diversity makes it difficult to create universally effective recommendations.

— Collaborative Filtering and Example:-

collaborative filtering is a technique used in recommendation systems where user preferences are analyzed based on shared behaviors with others. It works in two ways.

• User-based collaborative Filtering

If User A and User B have similar interests, recommendations for User A are based on what User B liked.

• Item-based collaborative Filtering:-

If two items are frequently bought/watched together, recommending one when the other is selected is effective.

Ex:-

Netflix suggests movies based on what users with similar viewing history have watched. If many users who watched *3 Idiots* also watched *Zindagi Na Milegi Dobara*.

— Role of clustering in Social Network Analysis.

Clustering helps group users with similar attributes in social networks, improving community detection, influence analysis and targeted marketing. It identifies friendship circles, political affiliations, or common interests.

for instance, in Twitter analysis, clustering can group users discussing similar topics, helping brands or policymakers understand audience sentiment and trends.



Q1:- What is the Random Forest Algorithm? Explain how it can be applied to classification problems. Write a program in R to implement a Random Forest classifier on a sample dataset and explain its output.

Ans:- Random Forest Algorithm and its Application in classification:-
Random forest is an ensemble learning algorithm that builds multiple decision trees and merges their result to produce a more accurate and stable prediction.
It works by creating a "forest" of decision trees where each tree is trained on a random subset of data and features.
The final prediction is determined by majority voting (for classification) or averaging (for regression).

— Application in classification problems:-
In classification tasks, Random forest is useful when dealing with dataset.

- Medical Diagnosis:- Predicting diseases based on patient symptoms.
 - Spam detection:- Classifying emails as Spam or not Spam.
 - Credit Risk Assessment:- Identifying risky loan applications.
- The algorithm reduces overfitting by averaging multiple trees and improves accuracy through random feature selection.

R Program for Random Forest Classification:-

```
install.packages("randomForest")  
library(randomForest)  
data(iris)  
set.seed(123)  
train_index <- sample(1:nrow(iris), 0.7 * nrow(iris))  
train_index <- iris[train_index,]  
test_data <- iris[-train_index,]  
  
rf_model <- randomForest(species ~ ., data = train_data,  
                           ntree = 100)
```




predictions $\leftarrow$ predict(rf-model, test_data)
conf_matrix $\leftarrow$ table(test_data\$Species, predictions)
accuracy $\leftarrow$ sum(diag(conf_matrix)) / sum(conf_matrix)

print(conf_matrix)

print(paste("Accuracy:", round(accuracy * 100, 2), "%"))

output:-

- The confusion matrix shows actual vs. predicted labels
- The accuracy value indicates how well the model classifies the test dataset.
- Higher ntrees values improve stability but increase computation time.

Random Forest provides high accuracy and robustness, making it a popular choice for classification tasks.

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